⌜Def. Let G be a group. Define

$$G^{(0)} = G \quad \text{and} \quad G^{(i+1)} = [G^{(i)}, G^{(i)}] = (G^{(i)})'$$

The descending chain

$$G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq G^{(2)} \trianglerighteq \cdots \qquad (i \in \mathbb{N})$$

is called the derived series of G.

⌜Thm. Let G be a group. Then,

G is solvable if and only if $G^{(r)} = 1$ for some $r \in \mathbb{N} \cup \{0\}$.

Proof. Suppose that G is solvable. That is, there is a subnormal series $1 = H_r \trianglelefteq H_{r-1} \trianglelefteq \cdots \trianglelefteq H_1 \trianglelefteq H_0 = G$ such that H_i / H_{i+1} is abelian for all $0 \leq i \leq r-1$.

Using induction for i, to show that $G^{(i)} \leq H_i$. If $i = 1$, then

$$G' = G^{(1)} \leq H_1 \iff G/H_1 \text{ is abelian}$$

So, it is true by the assumption. Now, suppose that for $i = k$, $G^{(k)} \leq H_k$. Then, $G^{(k+1)} = [G^{(k)}, G^{(k)}] \leq [H_k, H_k]$

and, since $H_{k+1} \trianglelefteq H_k$ with H_k / H_{k+1} is abelian, so $H_k' = [H_k, H_k] \leq H_{k+1}$, so $G^{(k+1)} \leq [H_k, H_k] \leq H_{k+1}$. So, by the induction, $G^{(r)} \leq H_r = 1$, so $G^{(r)} = 1$.

Conversely, suppose that $G^{(r)} = 1$ for some $r \in \mathbb{N}$. (If $r = 0$, then there are nothing to prove). Then, the series $\{G^{(i)}\}_{i=0}^{r}$ is the normal series such that

$$G^{(i)}/G^{(i+1)} \quad \text{for all } 0 \leq i \leq r-1,$$

Since $G^{(i+1)}$ is the derived subgroup of $G^{(i)}$, so proved ∎

⌈Lemma. If $\varphi : G \to H$ is a group Homomorphism, then

$$\varphi[G^{(k)}] = (\varphi[G])^{(k)} \text{ for all } k \in \mathbb{N}.$$

Proof since $\varphi([x,y]) = [\varphi(x), \varphi(y)]$, therefore

$$\varphi[G'] = \varphi[[G,G]] = [\varphi[G], \varphi[G]] = (\varphi[G])'.$$

and, the induction gives result. ☒ (엄밀하게는, 보여야 하지만, 귀찮다.)

⌈Prop. If G is solvable, then every subgroup and every quotient of G is solvable.

Proof. Let $r \in \mathbb{N}$ be a number such that $G^{(r)} = 1$.

Given a subgroup $H < G$, $H^{(i)} \leq G^{(i)}$ for all $0 \leq i \leq r$, so $H^{(r)} \leq G^{(r)} = 1$ ⌟

Given a normal subgroup $N \trianglelefteq G$, Consider the Canonical Projection $\pi : G \to G/N$.

Then, by the Lemma above,

$$(G/N)^{(r)} = (\pi[G])^{(r)} = \pi[G^{(r)}] = G^{(r)}N = \{N\}$$ So, Proved. ☒

⌈Prop. Let $N \trianglelefteq G$ be a normal subgroup.

If N and G/N are solvable, then G is solvable.

Proof Consider the Canonical Projection $\pi : G \to G/N$ and let $m, n \in \mathbb{N}$ such that

$$N^{(m)} = \{1\} \text{ and } (G/N)^{(n)} = \{1N\}$$

Since $(G/N)^{(n)} = G^{(n)}N = \{N\}$ means $G^{(n)} \leq N$,

$$G^{(m+n)} = (G^{(n)})^{(m)} \leq N^{(m)} = \{1\}$$

implies the result. ☒

⌈Debris.

$$(G/N)' = [G/N, G/N] = \langle [g_1N, g_2N] \mid g_1, g_2 \in G \rangle = \langle [g_1, g_2]N \mid g_1, g_2 \in G \rangle.$$

Meanwhile, since G' and N both are normal, $G'N$ is subgroup of G,
moreover, $N \trianglelefteq G'N$, so $G'N/N$ is well-defined quotient group.
Given $[g_1, g_2] \cdot nN$, $[g_1, g_2]nN = [g_1, g_2]N \in \langle [g_1, g_2]N \mid g_1, g_2 \in G \rangle$,
and the reverse inclusion is given by every element in the generated subgroup
is finite product of elements of $\{gN \mid g \in G'\}$, so

$$(G/N)' = G'N/N.$$

Inductively, $(G'N/N)' = (G'N)'N/N = G''N/N, \cdots, (G/N)^{(r)} = G^{(r)}N/N = N/N = 1.$

So Proved. ☒

⌜Solvabilty of the Symmetric group.

⌜S_3. Since $S_3 = \langle (1\,2), (1\,2\,3) \rangle$,

$$S_3' = [S_3, S_3] = \langle (1\,2)^{-1}(1\,2\,3)^{-1}(1\,2)(1\,2\,3) \rangle = \langle (1\,3\,2) \rangle = A_3.$$

And, $S_3/A_3 \cong C_2$ and $A_3/\{id\} \cong A_3 \cong C_3$ are abelian,

$$S_3 \rhd A_3 \rhd 1$$

Implies S_3 is solvable.

⌜A_4. Let $V = \{1, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\} \subseteq A_4$.
$A_4/V \cong C_3$, abelian, so $A_4' \leq V$. And, $A_4' = \{id\} \Leftrightarrow A_4$ is abelian
but $(1\,2\,3)\cdot(1\,2\,4) \neq (1\,2\,4)(1\,2\,3)$, so $\{id\} \lneq A_4' \leq V$.
Moreover, if $A_4' \neq V$, then either

$$A_4' = \langle (1\,2)(3\,4) \rangle \quad \text{or} \quad \langle (1\,3)(2\,4) \rangle \quad \text{or} \quad \langle (1\,4)(2\,3) \rangle$$

But, there are *not* normal; for instance, $(1\,2\,3)(1\,2)(3\,4)(3\,2\,1) = (1\,4) \notin \langle (1\,2)(3\,4) \rangle$,
So $A_4' = V$. Now,

$$A_4 \rhd V = A_4' \rhd 1$$

implies that A_4 is solvable.